

PS 1: Part I

Problem 1: Java programming basics

1-1)

fixed:

```
Java
import java.util.Scanner;

public class Problem1{
    /*
     * This static method should take an integer x and return:
     *   x * -1 if x is positive
     *   the unchanged value of x otherwise
     */
    public static int makeNeg(int x)
    {
        if (x > 0){
            x *= -1;
        }
        return x;
    }

    public static void main(String args)
    {
        Scanner console = new Scanner(System.in);

        System.out.print("Enter an integer x: ");
        int x = console.nextInt();

        System.out.println("makeNeg(x) = " + makeNeg(x));

        console.close();
    }
}
```

The file is also in the PS folder

1-2)

a) 48.0

b) 15c

c) 0

d) 11.0

e) 4.1

f) 0.0

g) 0.0

h) 0.2

i) 18e

j) 188

Problem 2: Conditional execution

2-1)

a)

lily

lavender

done

b)

hydrangea

marigold

done

c)

iris

daisy

done

d)

hydrangea

marigold

done

e)

rose

lavender

done

f)

daisy

done

2-2)

It is the "if(c < a)" statement on line 10.

This if statement cannot be executed because for line 10 to be run, first it has to pass through the if statements at lines 8 and 7, which state that a is smaller than b and b is smaller than c. If c is larger than b, that must mean that it is also larger than a, and as long as a isn't in a quantum state of uncertainty(which it isn't), it can't both be larger than and smaller than c.

3-1

x	y
10	2
54	2
54	3
54	2700

[illegible]

output (the lines printed by the program)

```
10 2
-6 -10
54 2
-2 2
54 3
-50 -54
54 2700
```

3-2)

Java

```
public static double pyramidVol(double l, double w, double h){
    double volume = (l*w*h)/3.0;
    return volume;
}
```

What is tri area and why does it take 3 variables?????

Problem 4: Loops

4-1)

```
for (int i = 0; i < 113; i++) {  
    System.out.println("I feel loopy!");  
}
```

4-2)

```
public static void mulFour(int n) {  
    for (int mul = n; mul >= 4; mul -= 4) {  
        System.out.println(mul);  
    }  
}
```

//While Loop Method:

```
public static void mulFourWhile(int n){  
    n -= n%4;  
    while(n>=4){  
        system.out.println(n);  
        n-=4;  
    }  
}
```

4-3)

```
for (int i = 0; i < 3; i++) {  
    for (int j = 0; j <= 2; j++) {  
        System.out.println(i + " " + j);  
    }  
    System.out.println("--");  
}  
//Fixed:  
for (int i = 0; i < 3; i++) {  
    for (int j = 0; j <= i; j++) {  
        System.out.println(i + " " + j);  
    }  
    System.out.println("--");  
}
```